操作系统原理实验报告

实验名称	授课老师	学生姓名	学生学号
编译内核/利用已有内核构建OS	张青	王健阳	21307261

一、实验要求

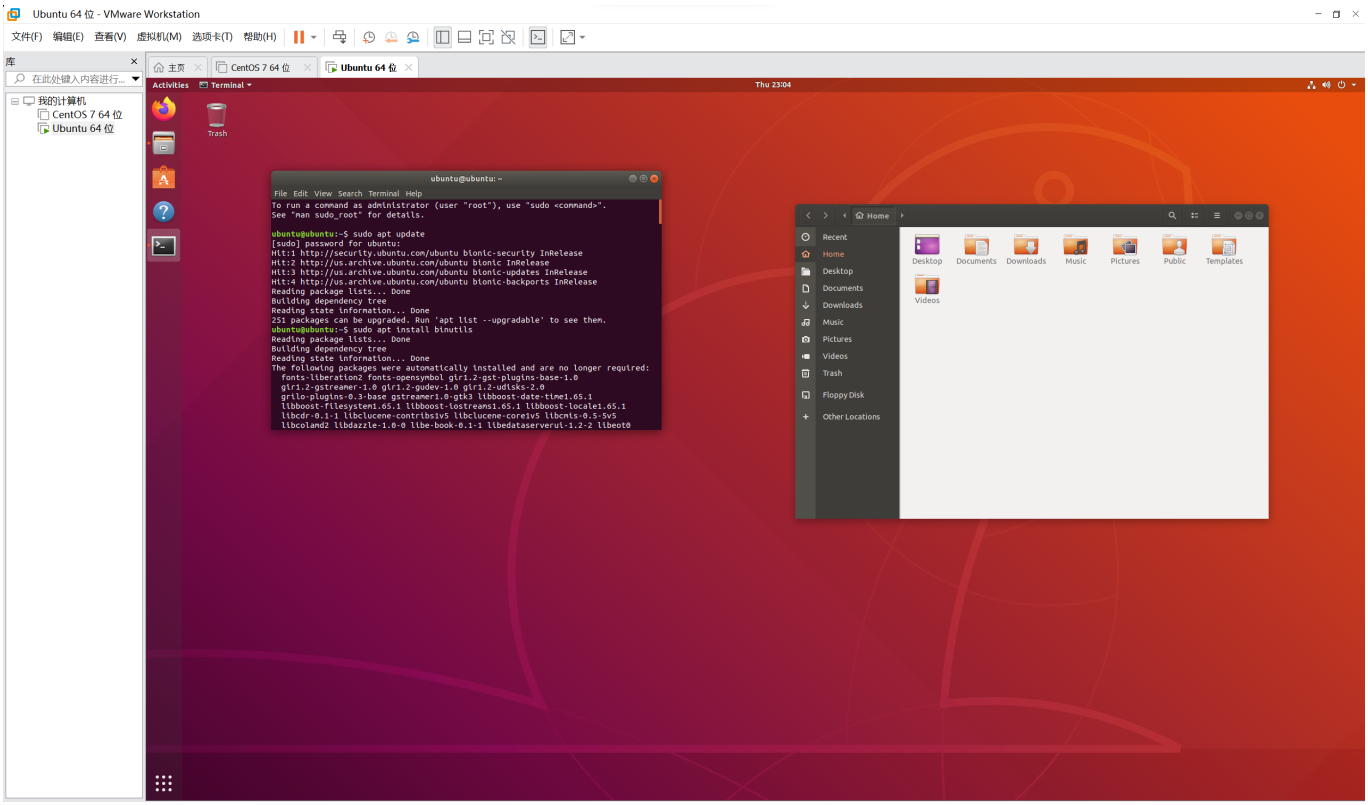
熟悉现有Linux内核的编译过程和启动过程，并在自行编译内核的基础上构建简单应用并启动；利用精简的Busybox工具集构建简单的OS，熟悉现代操作系统的构建过程。此外，熟悉编译环境、相关工具集，并能够实现内核远程调试；

1. 独立完成实验5个部份环境配置、编译Linux内核、Qemu启动内核并开启远程调试、制作Initramfs和编译并启动Busybox。
2. 编写实验报告、结合实验过程来谈谈你完成实验的思路和结果，最后需要提供实验的5个部份的程序运行截屏来证明你完成了实验。
3. 实验不限语言， C/C++/Rust都可以。
4. 实验不限平台， Windows、Linux和MacOS等都可以。
5. 实验不限CPU， ARM/Intel/Risc-V都可以。

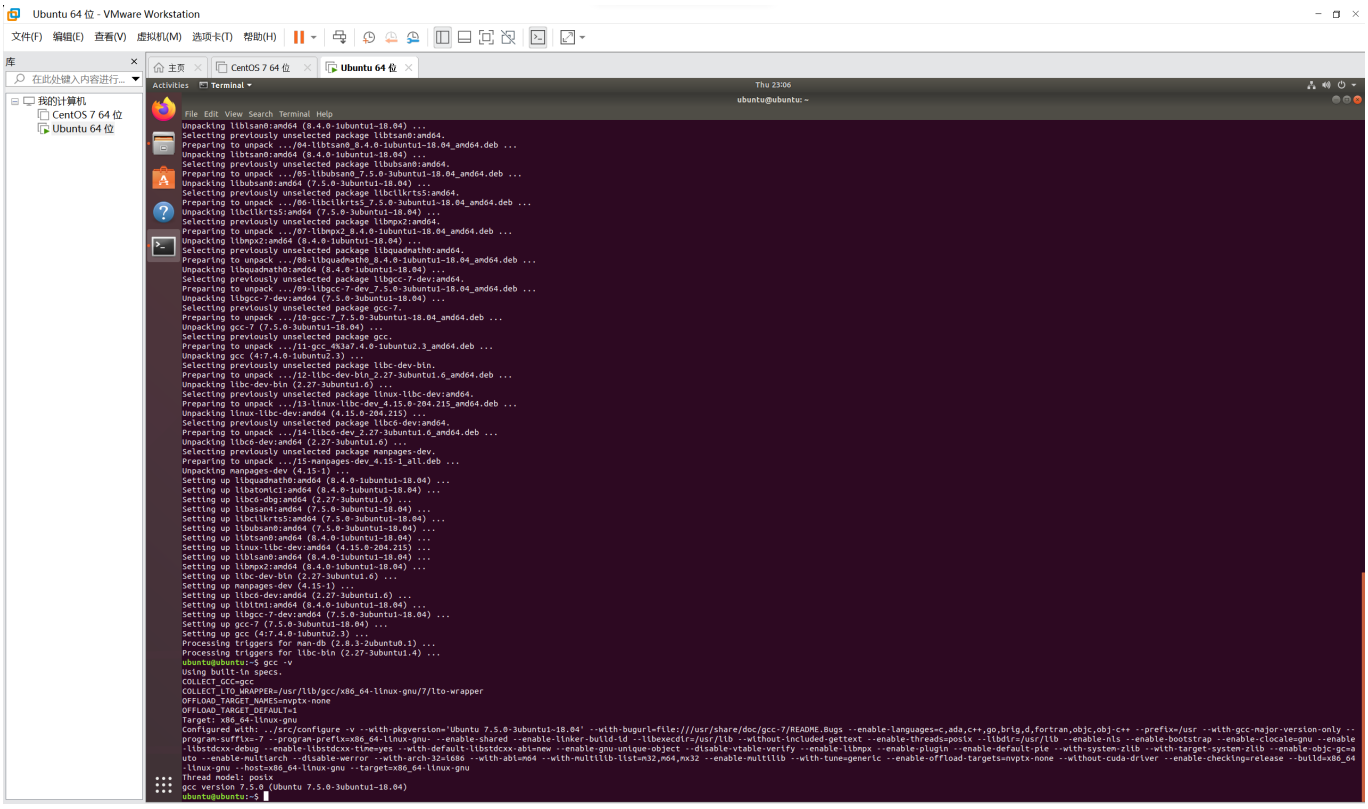
二、实验过程

1. 环境配置

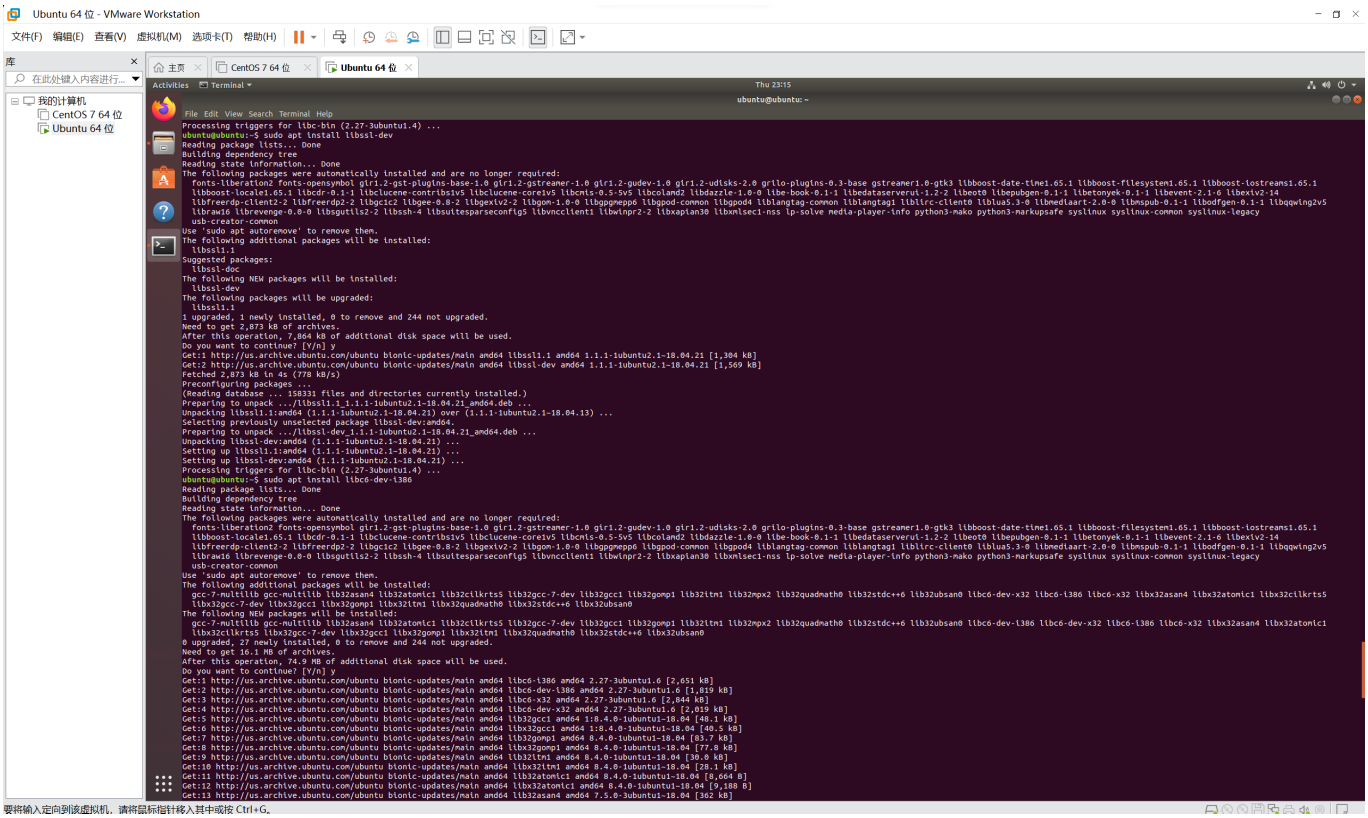
在电脑上搭建好Linux系统环境，利用VMware建好Ubuntu虚拟机并更新好apt



完成配置C++环境

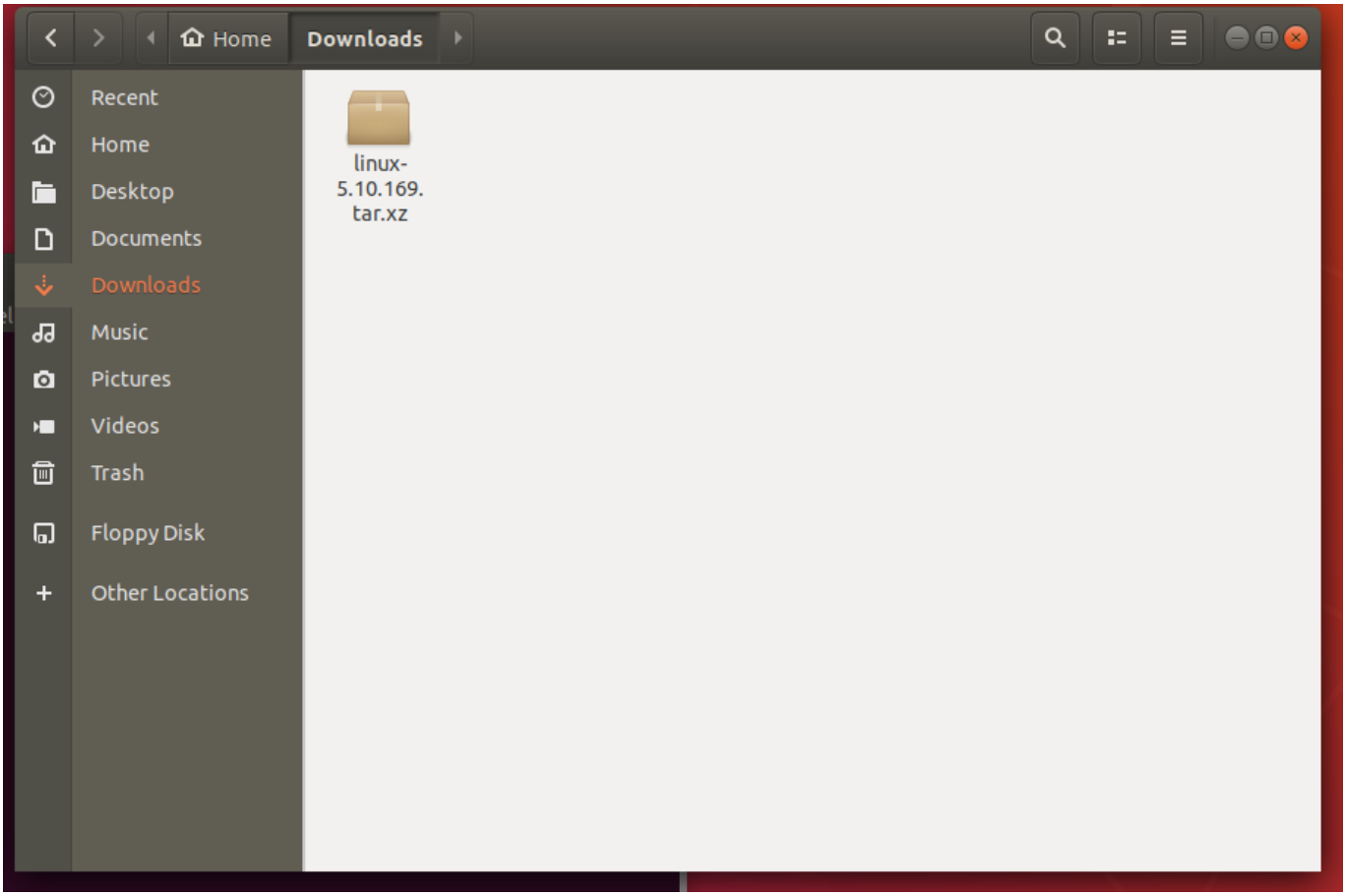


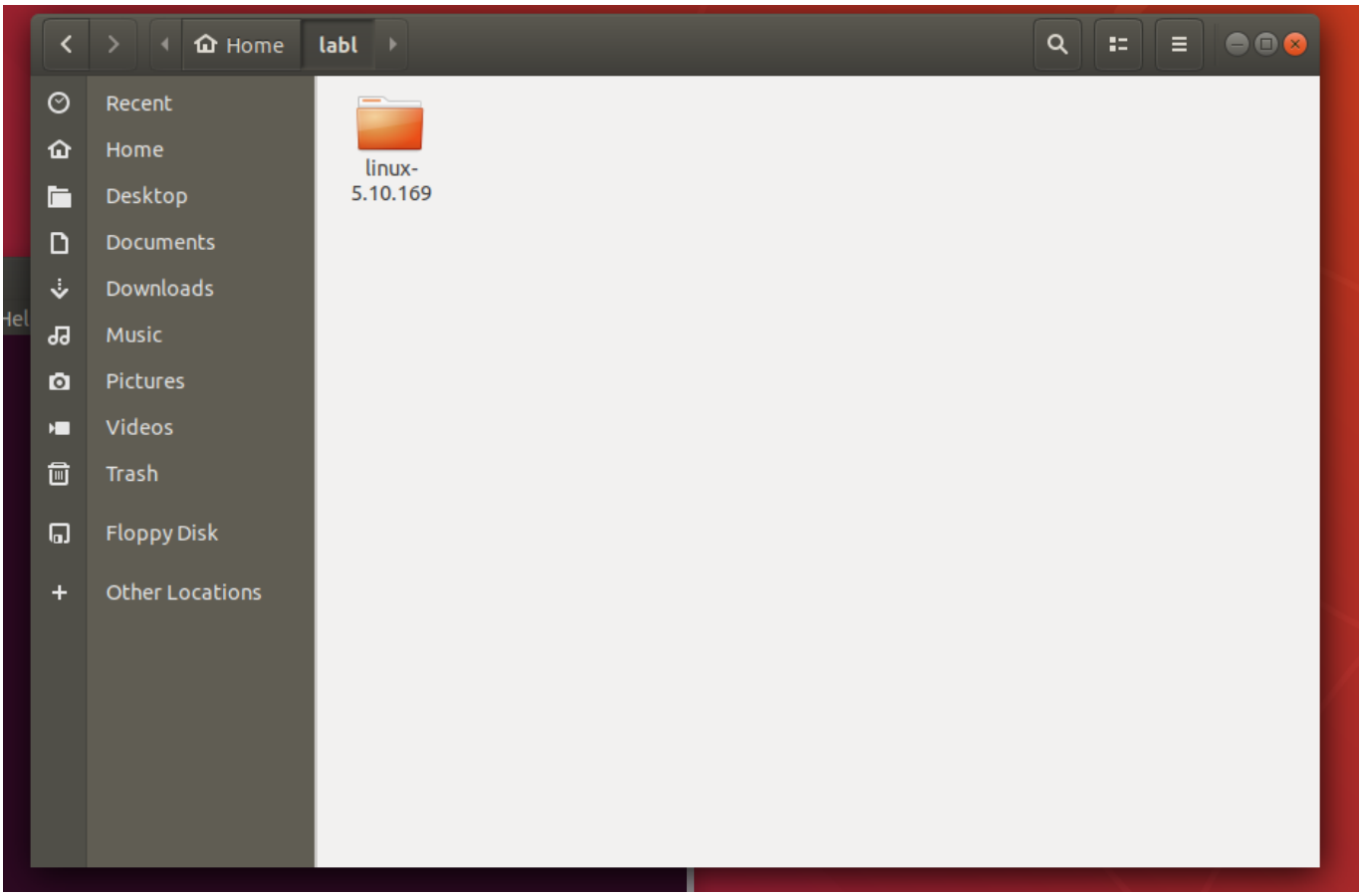
安装其他工具



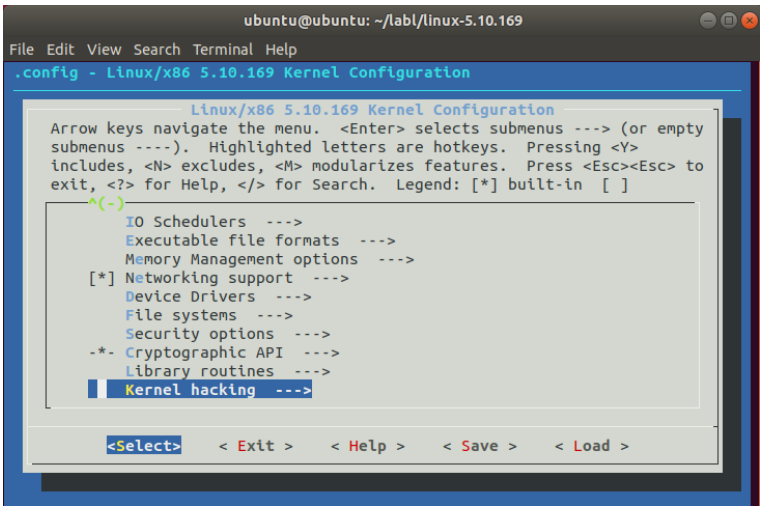
2. 编译 Linux 内核

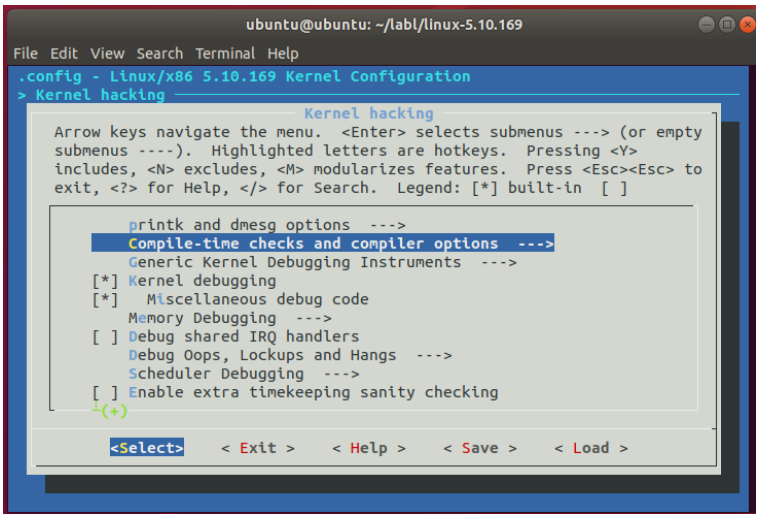
下载kernel 5.10 到文件夹labl（本该是lab1但打错了）

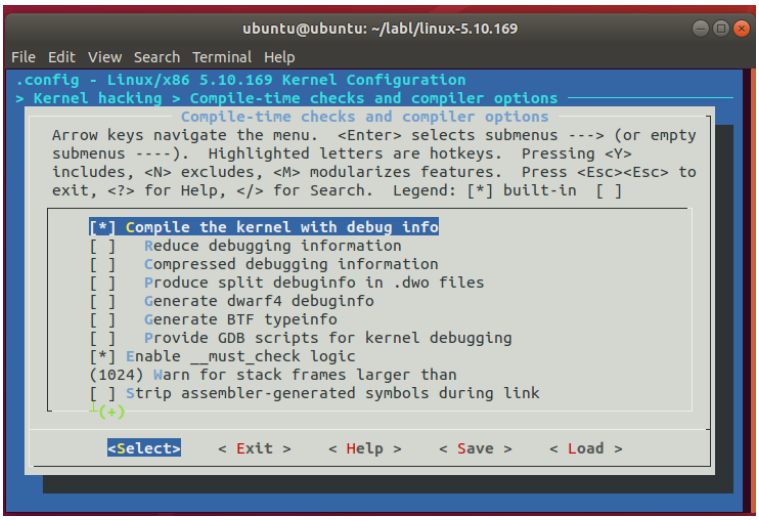



Home
labl
Recent
Home
Desktop
Documents
Downloads
Music
Pictures
Videos
Trash
Floppy Disk
Other Locations
linux-5.10.169

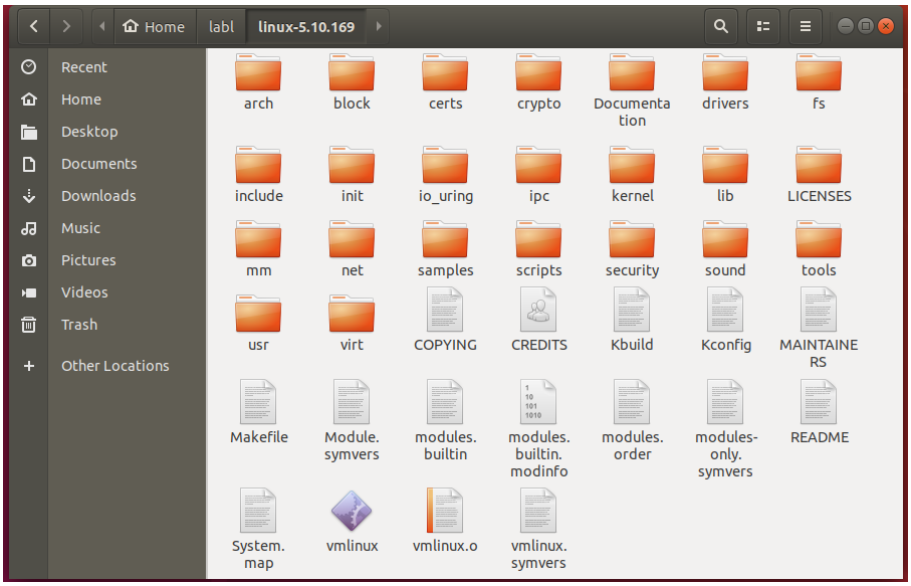
编译内核

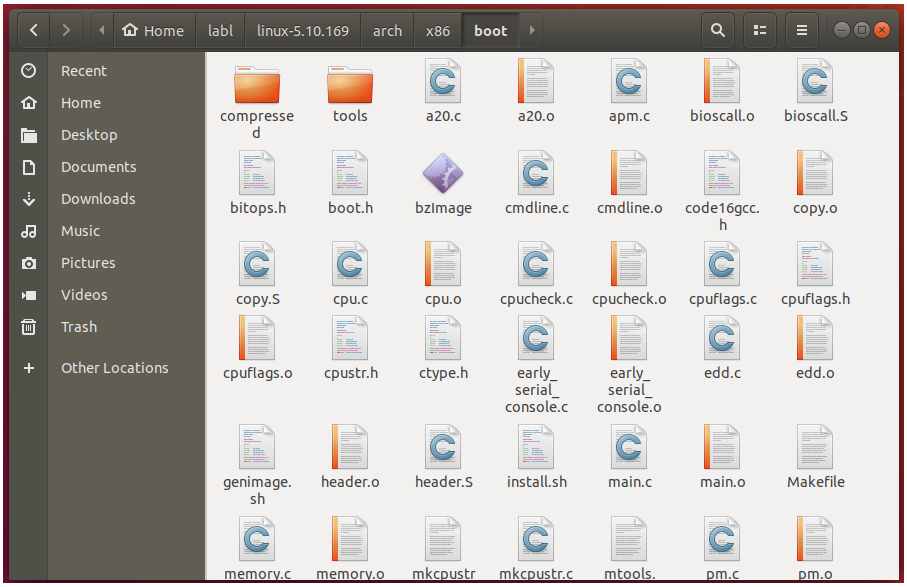

ubuntu@ubuntu: ~/labl/linux-5.10.169
File Edit View Search Terminal Help
.config - Linux/x86 5.10.169 Kernel Configuration
Linux/x86 5.10.169 Kernel Configuration
Arrow keys navigate the menu. <Enter> selects submenus ---> (or empty submenus ----). Highlighted letters are hotkeys. Pressing <Y> includes, <N> excludes, <M> modularizes features. Press <Esc><Esc> to exit, <?> for Help, </> for Search. Legend: [*] built-in [ ]
IO Schedulers --->
Executable file formats --->
Memory Management options --->
[*] Networking support --->
Device Drivers --->
File systems --->
Security options --->
-*- Cryptographic API --->
Library routines --->
Kernel hacking --->
<Select> < Exit > < Help > < Save > < Load >


ubuntu@ubuntu: ~/labl/linux-5.10.169
File Edit View Search Terminal Help
.config - Linux/x86 5.10.169 Kernel Configuration
> Kernel hacking
Kernel hacking
Arrow keys navigate the menu. <Enter> selects submenus ---> (or empty submenus ----). Highlighted letters are hotkeys. Pressing <Y> includes, <N> excludes, <M> modularizes features. Press <Esc><Esc> to exit, <?> for Help, </> for Search. Legend: [*] built-in [ ]
printk and dmesg options --->
Compile-time checks and compiler options --->
Generic Kernel Debugging Instruments --->
[*] Kernel debugging
[*] Miscellaneous debug code
Memory Debugging --->
[ ] Debug shared IRQ handlers
Debug Oops, Lockups and Hangs --->
Scheduler Debugging --->
[ ] Enable extra timekeeping sanity checking
<Select> < Exit > < Help > < Save > < Load >



编译后vmlinux与bzImage已成功生成





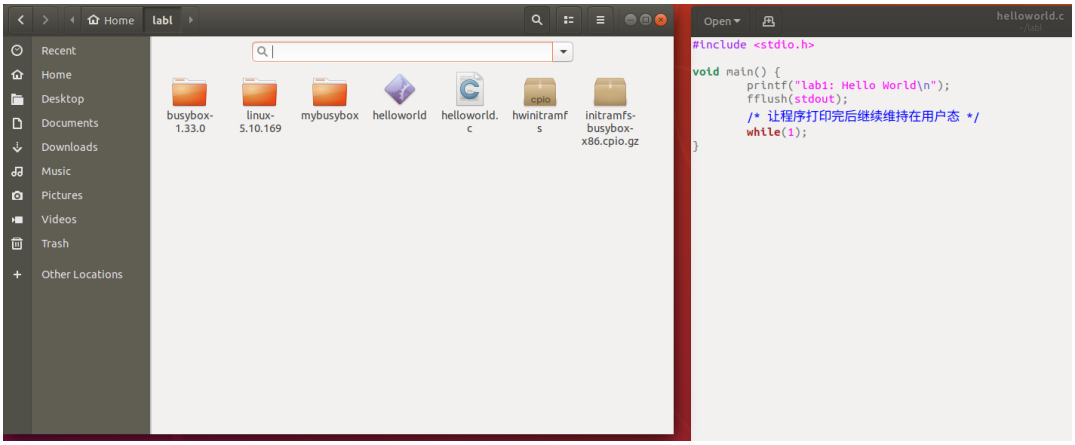
3. 启动内核并调试

此步骤是初次使用qemu启动内核并开启远程gdb调试，属于是一次试错的过程

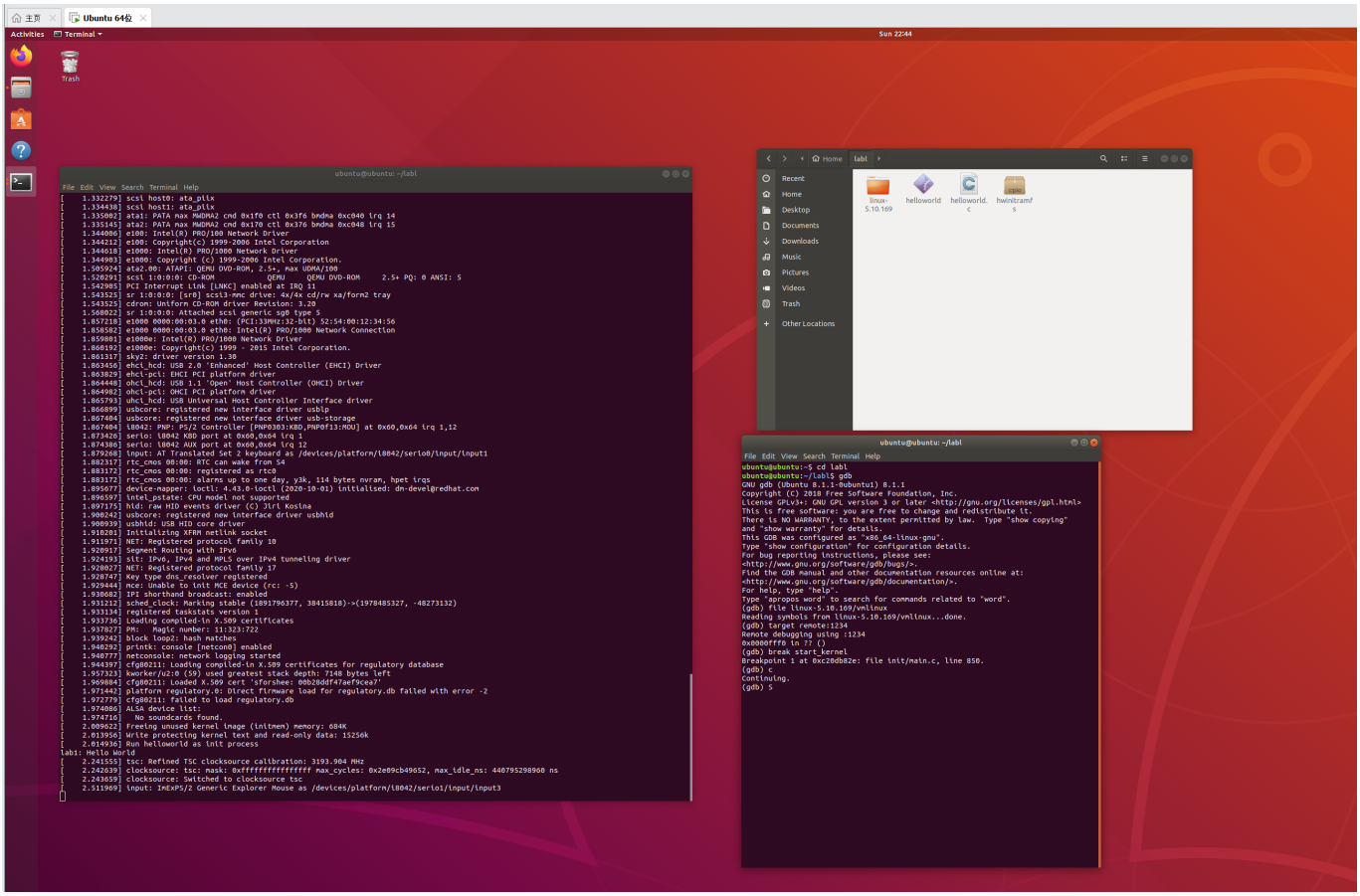
期间因为没有指定initrd文件，终端上没有任何输出结果

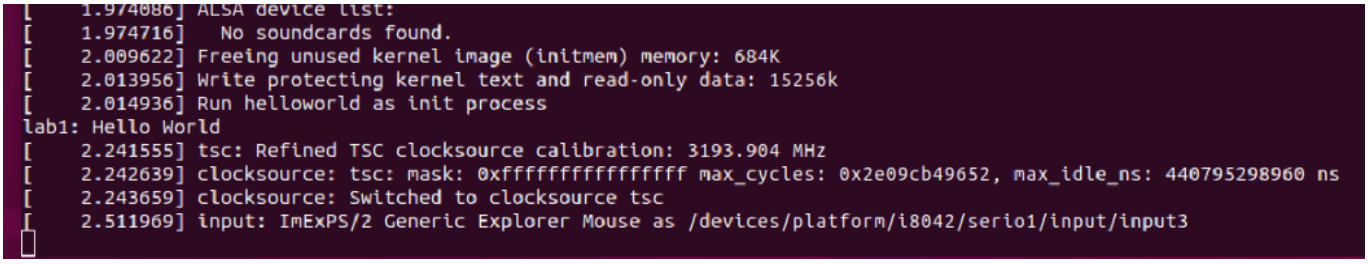
4. 制作Initramfs

编写helloworld程序，并用cpio打包Initramfs



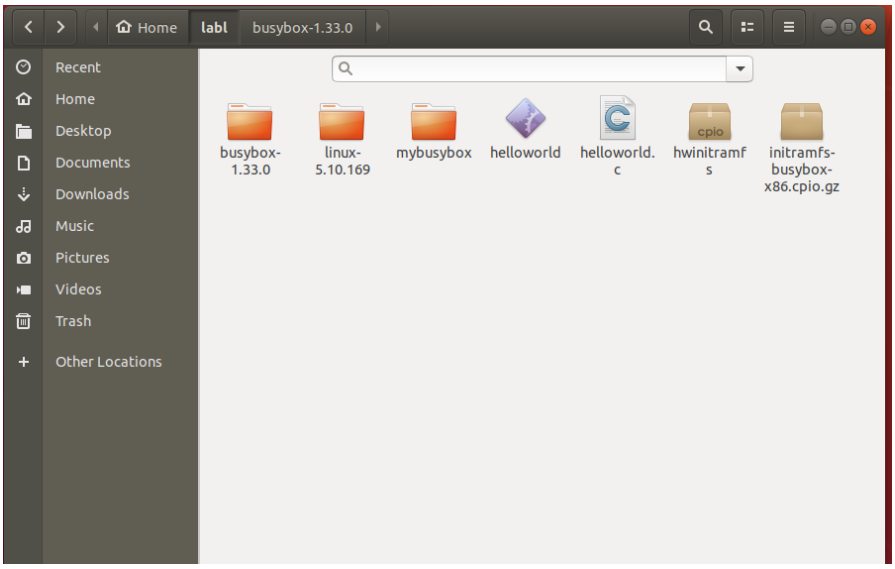
gdb调试



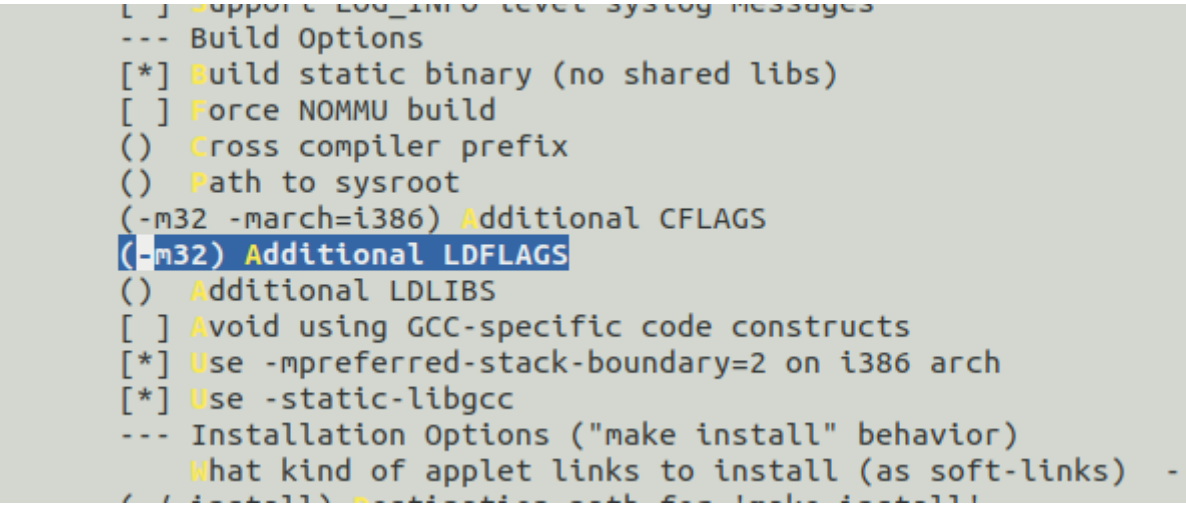


5. 编译并启动Busybox

下载busybox并解压到labl文件夹内



调整busybox设置



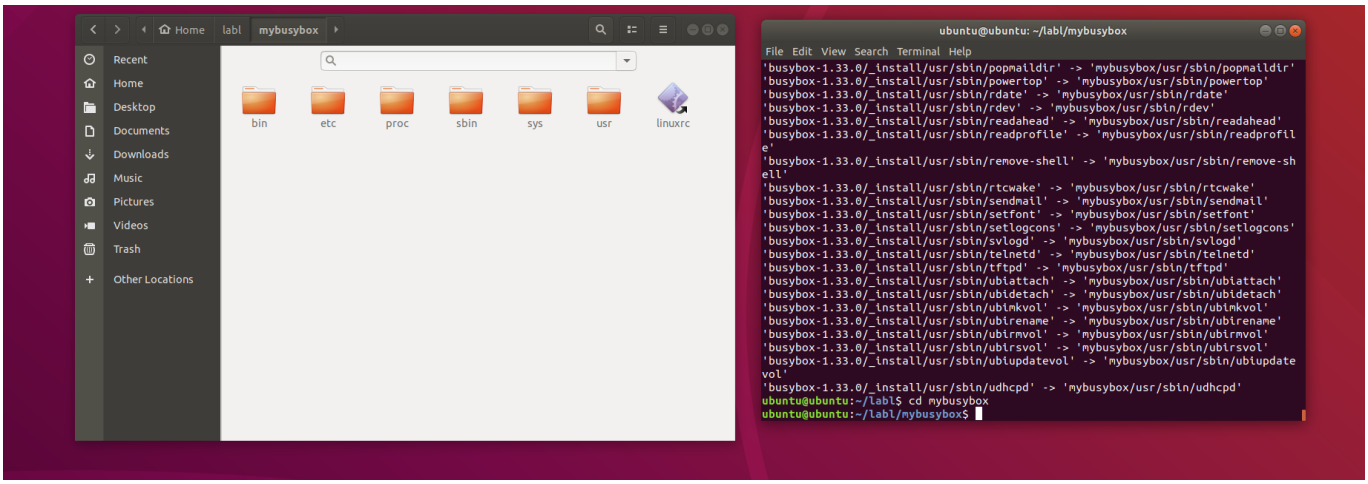
编译busybox

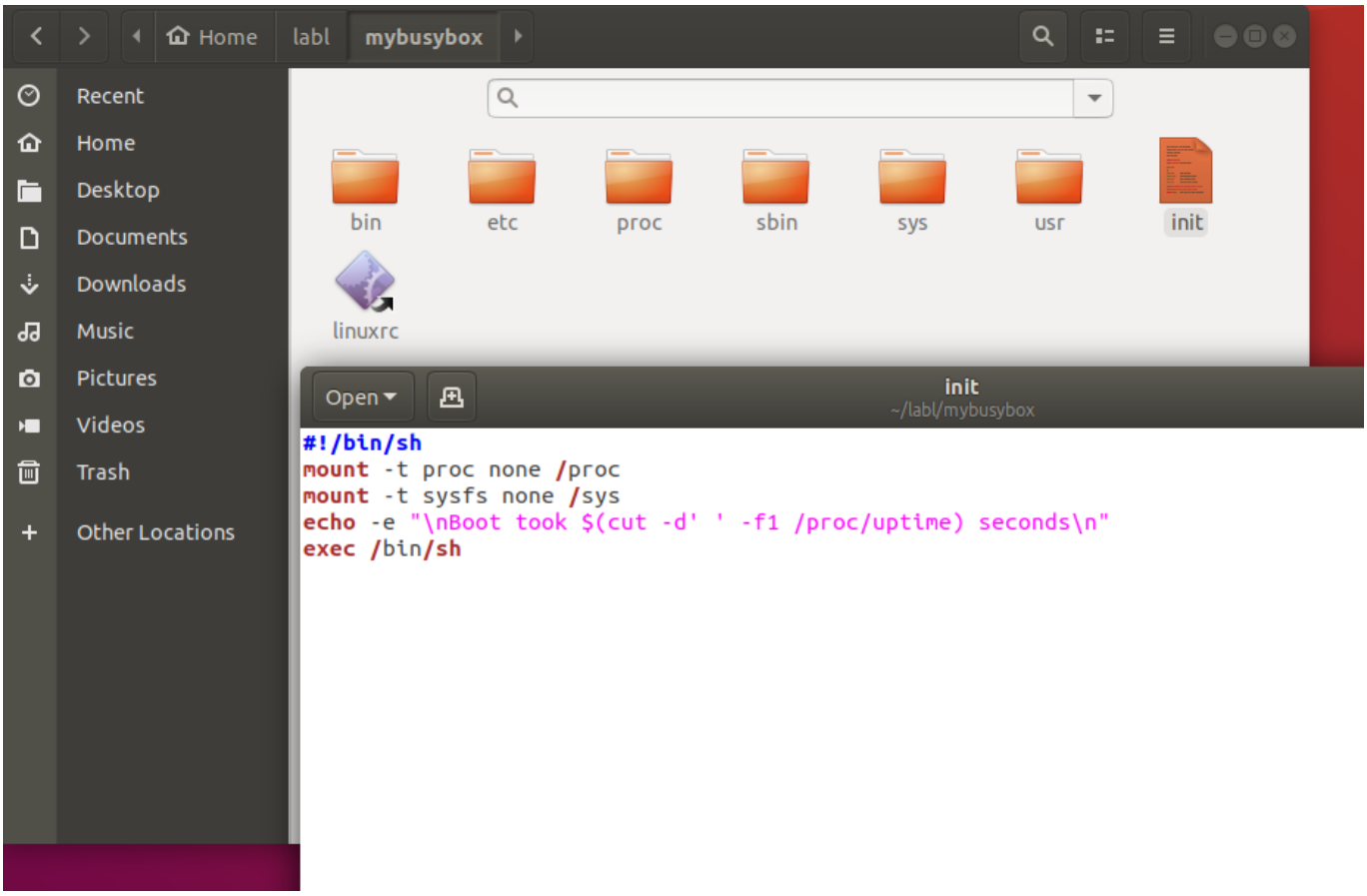
```
util-linux/unshare.c: In function 'unshare_main':
util-linux/unshare.c:295:4: warning: ignoring return value of 'read', declared with attribute warn_unused_result [-Wunused-result]
    read(fdp.rd, ns_ctx_list, 1); /* ...using bogus buffer */
    ^~~~~~~~~~~~~~~~~~~~~~~~~~~
  CC      libbb/percent_decode.o
  CC      libbb/perror_msg.o
  CC      coreutils/who.o
  CC      coreutils/whoami.o
  CC      networking/tls_pstm_montgomery_reduce.o
  CC      networking/tls_pstm_mul_comba.o
  AR      util-linux/lib.a
  CC      libbb/perror_nomsg.o
  CC      libbb/perror_nomsg_and_die.o
  CC      coreutils/yes.o
  CC      libbb/pidfile.o
  CC      libbb/platform.o
  CC      libbb/print_flags.o
  CC      libbb/print_numbered_lines.o
  CC      networking/tls_pstm_sqr_comba.o
  CC      networking/tls_rsa.o
  AR      coreutils/lib.a
  CC      libbb/printable.o
  CC      libbb/printable_string.o
  CC      libbb/process_escape_sequence.o
  CC      libbb/procps.o
  CC      libbb/progress.o
  CC      networking/traceroute.o
  CC      networking/tunctl.o
  CC      networking/vconfig.o
  CC      libbb/ptr_to_globals.o
  CC      libbb/pw_encrypt.o
  CC      libbb/read.o
  CC      libbb/read_key.o
  CC      libbb/read_printf.o
  CC      networking/wget.o
  CC      networking/whois.o
  CC      libbb/recursive_action.o
  CC      libbb/remove_file.o
  CC      libbb/replace.o
networking/wget.c: In function 'retrieve_file_data':
networking/wget.c:1057:5: warning: ignoring return value of 'ftruncate', declared with attribute warn_unused_result [-Wunused-result]
    ftruncate(G.output_fd, pos);
    ^~~~~~~~~~~~~~~~~~~~~~~~~~~
  CC      libbb/rtc.o
  CC      networking/zcip.o
  CC      libbb/run_shell.o
  CC      libbb/safe_gethostname.o
  CC      libbb/safe_poll.o
  CC      libbb/safe_strncpy.o
  CC      libbb/safe_write.o
  CC      libbb/securetty.o
  CC      libbb/setup_environment.o
  CC      libbb/signals.o
  CC      libbb/simplify_path.o
  CC      libbb/single_argv.o
  CC      libbb/skip_whitespace.o
  CC      libbb/speed_table.o
  CC      libbb/str_tolower.o
  AR      networking/lib.a
  CC      libbb/strrstr.o
  CC      libbb/sysconf.o
  CC      libbb/time.o
  CC      libbb/trim.o
  CC      libbb/u_signal_names.o
  CC      libbb/ubi.o
  CC      libbb/udp_io.o
  CC      libbb/unicode.o
  CC      libbb/update_passwd.o
  CC      libbb/utmp.o
  CC      libbb/verror_msg.o
  CC      libbb/uuencode.o
  CC      libbb/vfork_daemon_rexec.o
  CC      libbb/warn_ignoring_args.o
  CC      libbb/wfopen.o
libbb/update_passwd.c: In function 'update_passwd':
libbb/update_passwd.c:153:3: warning: ignoring return value of 'fchown', declared with attribute warn_unused_result [-Wunused-result]
   fchown(new_fd, sb.st_uid, sb.st_gid);
   ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
  CC      libbb/wfopen_input.o
  CC      libbb/xatonum.o
  CC      libbb/write.o
  CC      libbb/xconnect.o
  CC      libbb/xfunc_die.o
  CC      libbb/xfuncs.o
  CC      libbb/xfuncs_printf.o
  CC      libbb/xgetcwd.o
  CC      libbb/xgethostbyname.o
  CC      libbb/xreadlink.o
  CC      libbb/xrealloc_vector.o
  CC      libbb/xregcomp.o
  AR      libbb/lib.a
  LINK    busybox_unstripped
Static linking against glibc, can't use --gc-sections
Trying libraries: crypt m resolv rt
 Library crypt is not needed, excluding it
 Library m is needed, can't exclude it (yet)
 Library resolv is needed, can't exclude it (yet)
 Library rt is not needed, excluding it
 Library m is needed, can't exclude it (yet)
 Library resolv is needed, can't exclude it (yet)
Final link with: m resolv
ubuntu@ubuntu:~/labl/busybox-1.33.0$ s
```

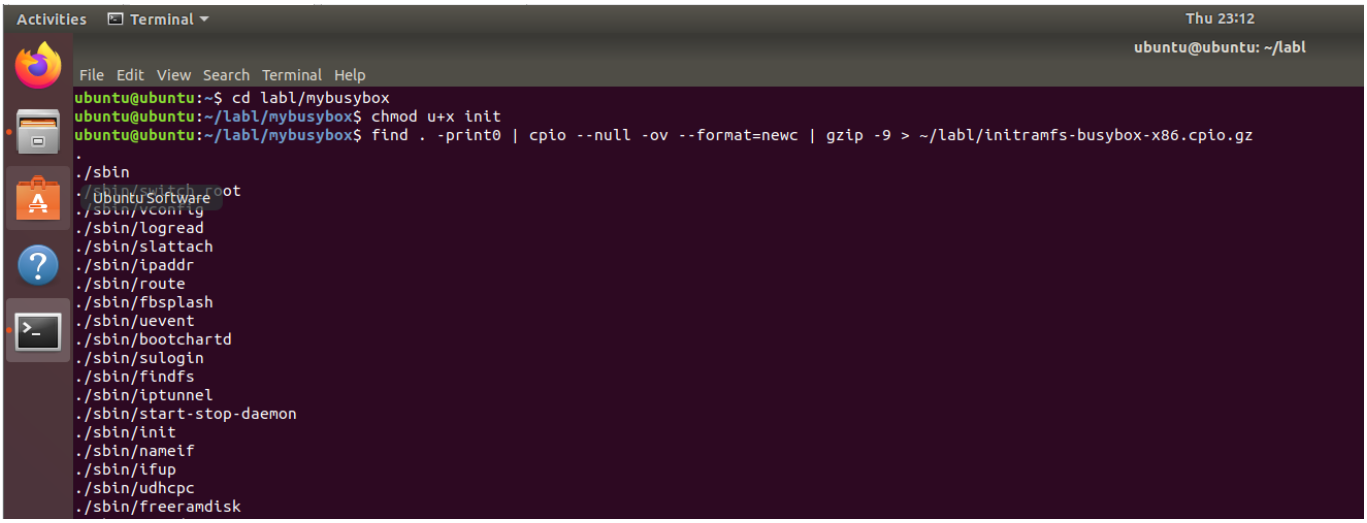
make install

```
./_install//usr/sbin/remove-shell -> ../../bin/busybox
./_install//usr/sbin/rtcwake -> ../../bin/busybox
./_install//usr/sbin/sendmail -> ../../bin/busybox
./_install//usr/sbin/setfont -> ../../bin/busybox
./_install//usr/sbin/setlogcons -> ../../bin/busybox
./_install//usr/sbin/svlogd -> ../../bin/busybox
./_install//usr/sbin/telnetd -> ../../bin/busybox
./_install//usr/sbin/tftpd -> ../../bin/busybox
./_install//usr/sbin/ubiattach -> ../../bin/busybox
./_install//usr/sbin/ubidetach -> ../../bin/busybox
./_install//usr/sbin/ubimkvol -> ../../bin/busybox
./_install//usr/sbin/ubirename -> ../../bin/busybox
./_install//usr/sbin/ubirmvol -> ../../bin/busybox
./_install//usr/sbin/ubirsvol -> ../../bin/busybox
./_install//usr/sbin/ubiupdatevol -> ../../bin/busybox
./_install//usr/sbin/udhcpd -> ../../bin/busybox


---------------------------------------------------
You will probably need to make your busybox binary
setuid root to ensure all configured applets will
work properly.
---------------------------------------------------

ubuntu@ubuntu:~/labl/busybox-1.33.0$ S
```

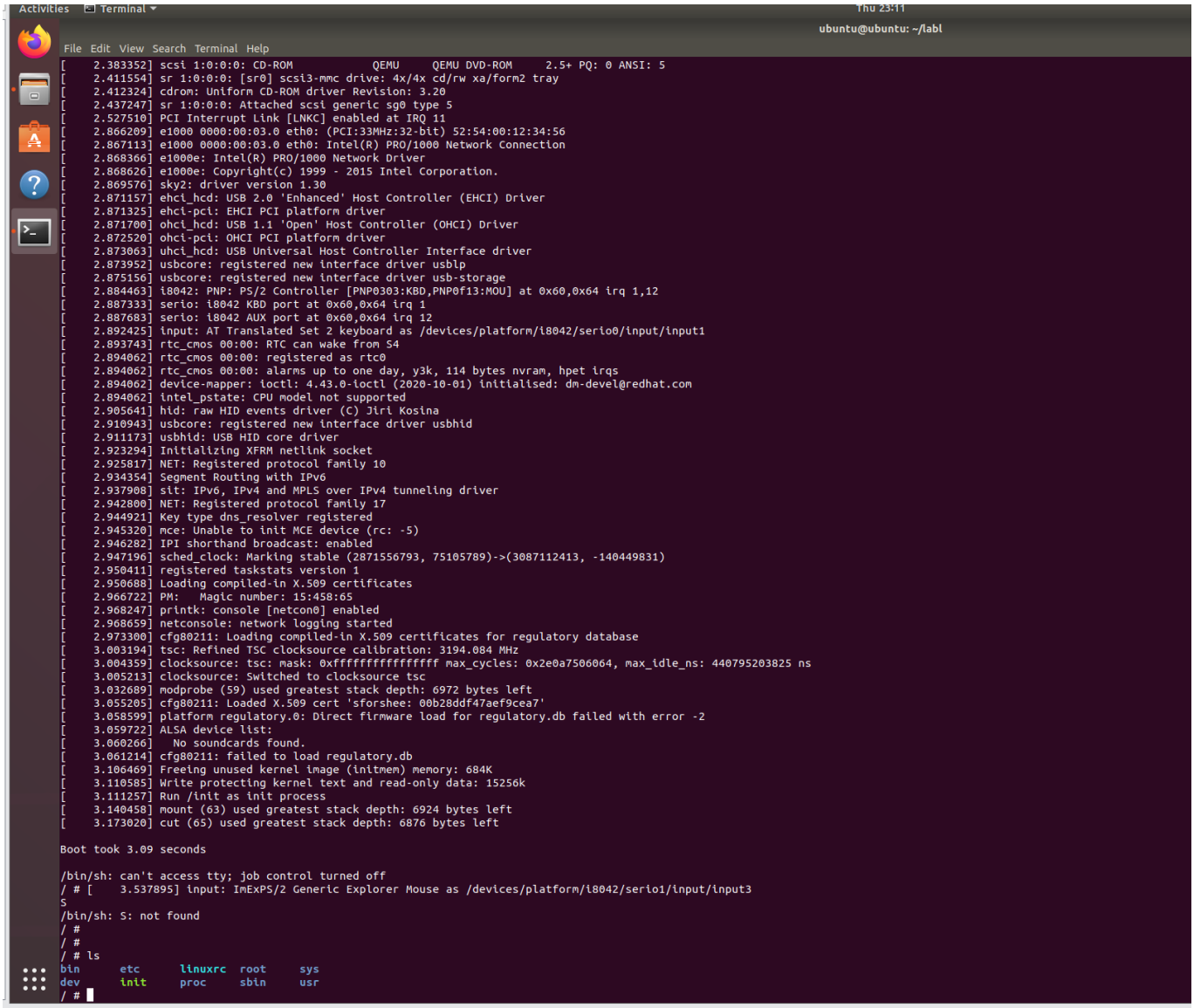
复制到mybusybox并制作Initramfs




Home
labl
mybusybox
Recent
Home
Desktop
Documents
Downloads
Music
Pictures
Videos
Trash
Other Locations
bin
etc
proc
sbin
sys
usr
init
linuxrc
Open
init
~/labl/mybusybox
#!/bin/sh
mount -t proc none /proc
mount -t sysfs none /sys
echo -e "\nBoot took $(cut -d' ' -f1 /proc/uptime) seconds\n"
exec /bin/sh


Activities
Terminal
Thu 23:12
ubuntu@ubuntu: ~/labl
File Edit View Search Terminal Help
ubuntu@ubuntu:~$ cd labl/mybusybox
ubuntu@ubuntu:~/labl/mybusybox$ chmod u+x init
ubuntu@ubuntu:~/labl/mybusybox$ find . -print0 | cpio --null -ov --format=newc | gzip -9 > ~/labl/initramfs-busybox-x86.cpio.gz
.
./sbin
./sbin/switch_root
./sbin/vconfig
./sbin/logread
./sbin/slattach
./sbin/ipaddr
./sbin/route
./sbin/fbsplash
./sbin/uevent
./sbin/bootchartd
./sbin/sulogin
./sbin/findfs
./sbin/iptunnel
./sbin/start-stop-daemon
./sbin/init
./sbin/nameif
./sbin/ifup
./sbin/udhcpc
./sbin/freeramdisk
Ubuntu Software

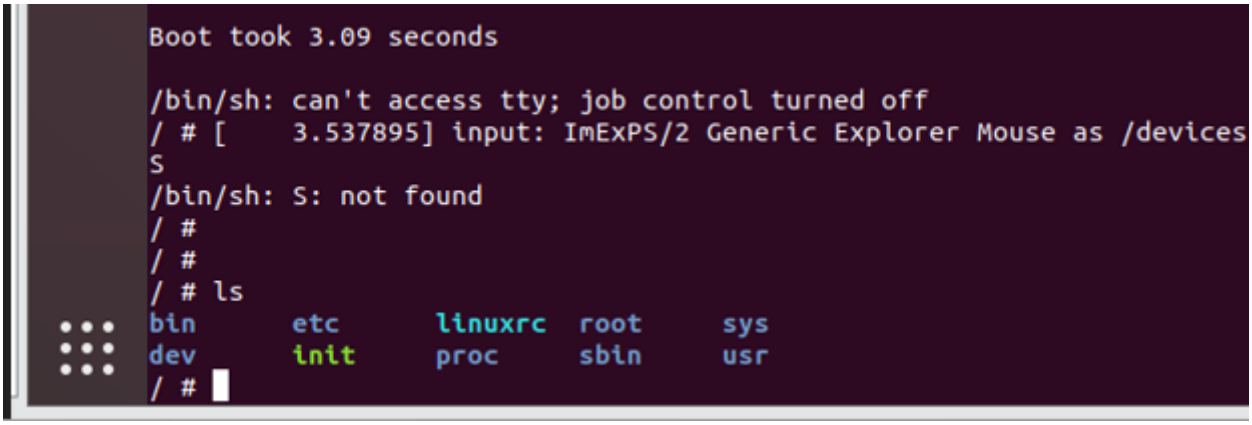
加载busybox



三、关键代码

此次为第一次实验，所有涉及的代码均在pdf中给出

四、实验结果



最后目标文件夹内容与pdf截图展示的一致，故实验成功

五、总结

这次实验是我第一次较为深入的接触Linux操作系统，初步了解到了Linux的一些基本命令。在实验期间，虽然全部指令都在pdf中有所给出，但实操时还是难免遇到点阻碍。例如需要输入较长命令的时候，一开始我往往会缺少或添加了一些空格，导致命令无法执行（事实上pdf上确实是有一条指令多了一个空格，直接粘贴不太行）。还有就是虚拟机的配置问题。一开始我给我的虚拟机只配了1G内存，结果导致编译内核的时候卡顿闪退，但是该到了4G之后就明显流畅多了。（编译内核的过程比我之前想象的要更吃内存）除此之外，就没什么别的阻碍了。最后就依次按照步骤，利用精简工具集busybox构建了一个简单的OS。熟悉了一下编译环境以及相关工具集。